

EF model - MIDA

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/EF_MIDA")
data <- read.csv("MIDA.csv")

# Assign factors
data$ROI <- factor(data$ROI)
data$Region <- factor(data$Region)
data$StimType <- factor(data$StimType)
data$Stim <- factor(data$Stim)
```

Plot data - Figure 1e

```
## TI Plot Cortical ROIs and Hippocampus
TIfields_cortex <- subset(data, Region=="Cortex" & StimType=="TI")

p.TI.field_cortex <- ggplot(TIfields_cortex, aes(x = ROI, y = Median)) +
  geom_bar(stat="identity", fill=c("#3b37a3", "#3b37a3", "#3b37a3")) +
  geom_errorbar(aes(ymin=Median-Std, ymax=Median+Std), width=0.3, size=.6,
    position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, .5)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
```

```

    panel.background = element_blank() +
    theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Env"

## Hippocampus all
TIfields_hipp <- subset(data, Region=="Hippocampus" & StimType=="TI")

## Median weighted by volume
Ant_w = (TIfields_hipp$Median[TIfields_hipp$ROI=="Ant"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI=="Ant"]
Mid_w = (TIfields_hipp$Median[TIfields_hipp$ROI=="Mid"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI=="Mid"]
Post_w = (TIfields_hipp$Median[TIfields_hipp$ROI=="Post"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI=="Post"]

## Std weighted by volume
Ant_w_sd = (TIfields_hipp$Std[TIfields_hipp$ROI=="Ant"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI=="Ant"]
Mid_w_sd = (TIfields_hipp$Std[TIfields_hipp$ROI=="Mid"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI=="Mid"]
Post_w_sd = (TIfields_hipp$Std[TIfields_hipp$ROI=="Post"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI=="Post"]

Hipp_t = Ant_w + Mid_w + Post_w
Hipp_sd = Ant_w_sd + Mid_w_sd + Post_w_sd
Region <- "Hipp"
Median <- Hipp_t
SD <- Hipp_sd
tVolBrain = sum(TIfields_hipp$VolBrain)

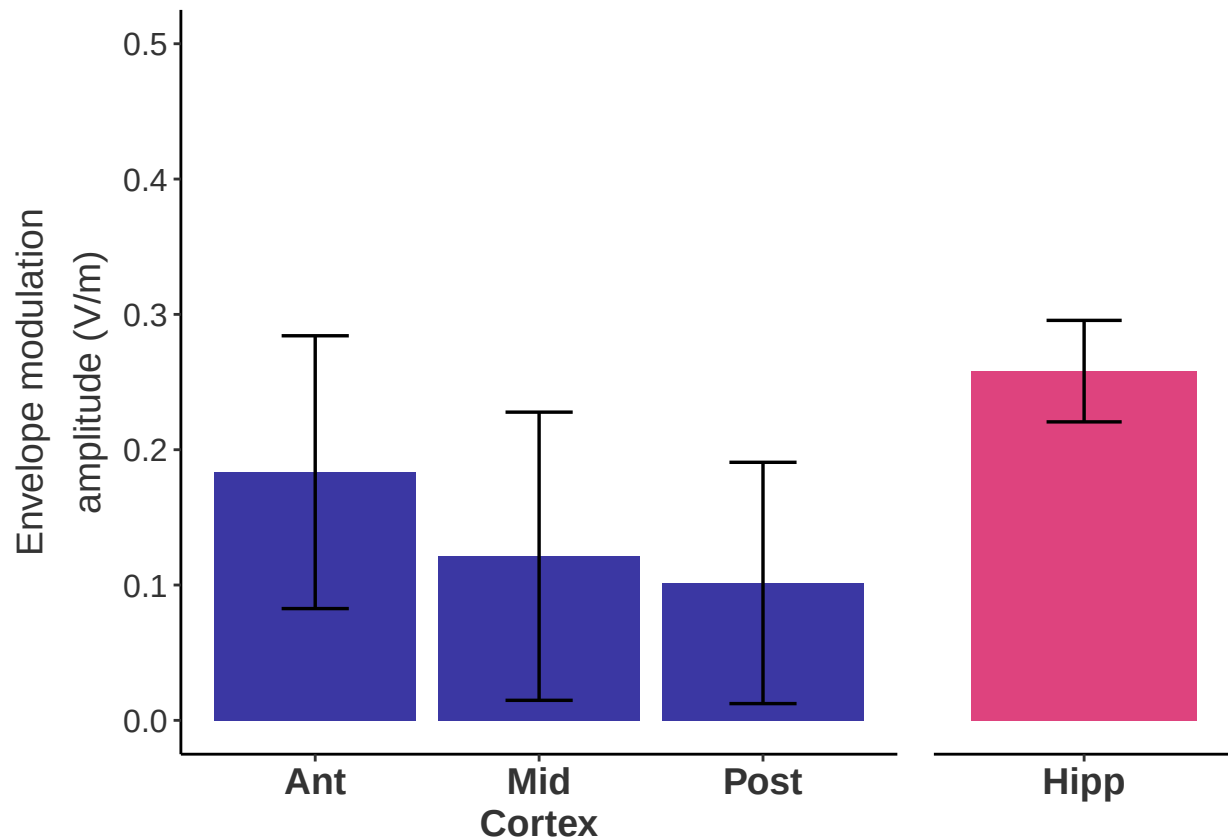
Thipp <- data.frame(Region, Median, SD)

p.TI.field_hipp <- ggplot(Thipp , aes(x = Region, y = Median)) +
  geom_bar(stat="identity", fill=c("#de437e")) +
  geom_errorbar(aes(ymin=Median-SD, ymax=Median+SD), width=0.3, size=.6, position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, .5)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_blank(),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_blank(),
    #axis.ticks.y=element_blank(),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank() +
    theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
    theme(strip.text.x = element_text(size = 14),

```

```
strip.text.y = element_text(size = 14), legend.position = "none") + xlab("") + easy_remove_y,
```

```
Fields_cortex_hipp <- grid.arrange(  
  arrangeGrob(p.TI.field_cortex, p.TI.field_hipp, nrow=1, ncol=2 ,widths=c(1,0.37)))
```



```
## ----- Normlaise to hippocampus -----  
fields_cortex <- TIfields_cortex %>% dplyr::select(ROI, Median, Std)  
fields_cortex <- rename(fields_cortex,  
  Region = ROI,  
  SD = Std)  
  
TIfields <- rbind(Thipp, fields_cortex)  
  
TIfields$Median[TIfields$Region=="Hipp"]
```

```
## [1] 0.2580584
```

```
norm <- ddply(TIfields, c("Region"), summarise,  
  Mnorm = (Median*100)/(TIfields$Median[TIfields$Region=="Hipp"]),  
  SDnorm = (SD)/(TIfields$Median[TIfields$Region=="Hipp"])*100  
)
```

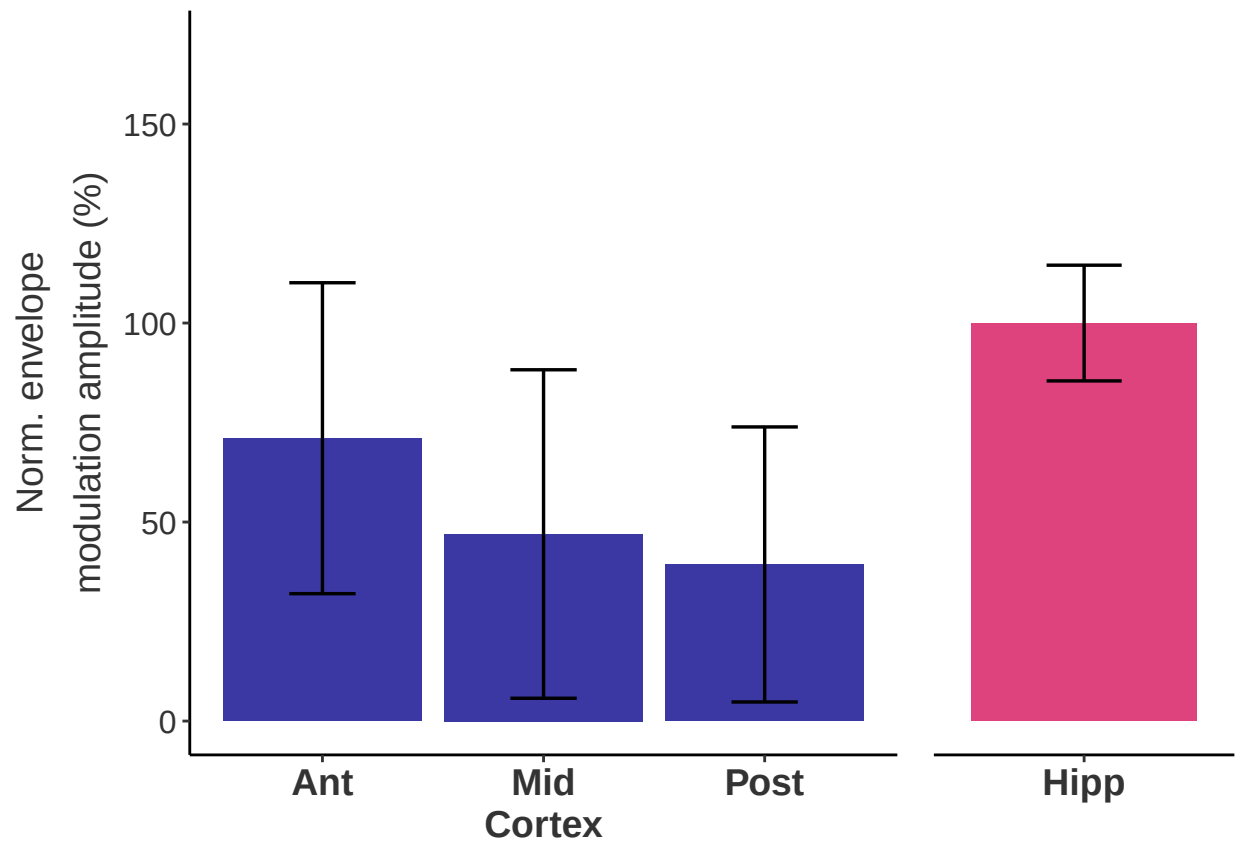
```

norm_cortex <- subset(norm, Region!="Hipp")
p.TI.field_cortex_norm <- ggplot(norm_cortex, aes(x = Region, y = Mnorm)) +
  geom_bar(stat="identity", fill=c("#3b37a3", "#3b37a3", "#3b37a3")) +
  geom_errorbar(aes(ymin=Mnorm-SDnorm, ymax=Mnorm+SDnorm), width=0.3, size=.6,
    position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, 170)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Norm", "Mn")))

norm_hipp <- subset(norm, Region=="Hipp")
p.TI.field_hipp_norm <- ggplot(norm_hipp, aes(x = Region, y = Mnorm)) +
  geom_bar(stat="identity", fill=c("#de437e")) +
  geom_errorbar(aes(ymin=Mnorm-SDnorm, ymax=Mnorm+SDnorm), width=0.3, size=.6, position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, 170)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_blank(),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_blank(),
    #axis.ticks.y=element_blank(),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 14), legend.position = "none") + xlab("") + easy_remove_y

Fields_cortex_hipp_norm <- grid.arrange(
  arrangeGrob(p.TI.field_cortex_norm, p.TI.field_hipp_norm, nrow=1, ncol=2, widths=c(1,0.37)))

```



```
ggsave(Fields_cortex_hipp_norm, file="Figure_1ei.eps", width = 9, height = 8, dpi = 300, units = "cm", d
```

Plot data - Figure 1e

```
## Absolute Amplitude
p.HF.field_cortex <- ggplot(TIfields_cortex, aes(x = ROI, y = MedianTEf)) +
  geom_bar(stat="identity", fill=c("#3b37a3", "#3b37a3", "#3b37a3")) +
  geom_errorbar(aes(ymin=MedianTEf-StdTEf, ymax=MedianTEf+StdTEf), width=0.3, size=.6,
    position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, .5)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "plain"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
```

```

strip.text.y = element_text(size = 14), legend.position = "none") + xlab("Cortex") + ylab(exp

## Hippocampus all
## Median weighted by volume
Ant_HF = (TIfields_hipp$MedianTEf[TIfields_hipp$ROI=="Ant"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI==
Mid_HF = (TIfields_hipp$MedianTEf[TIfields_hipp$ROI=="Mid"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI==
Post_HF = (TIfields_hipp$MedianTEf[TIfields_hipp$ROI=="Post"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI=

## Std weighted by volume
Ant_HF_sd = (TIfields_hipp$StdTEf[TIfields_hipp$ROI=="Ant"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI==
Mid_HF_sd = (TIfields_hipp$StdTEf[TIfields_hipp$ROI=="Mid"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI==
Post_HF_sd = (TIfields_hipp$StdTEf[TIfields_hipp$ROI=="Post"] * TIfields_hipp$VolBrain[TIfields_hipp$ROI=

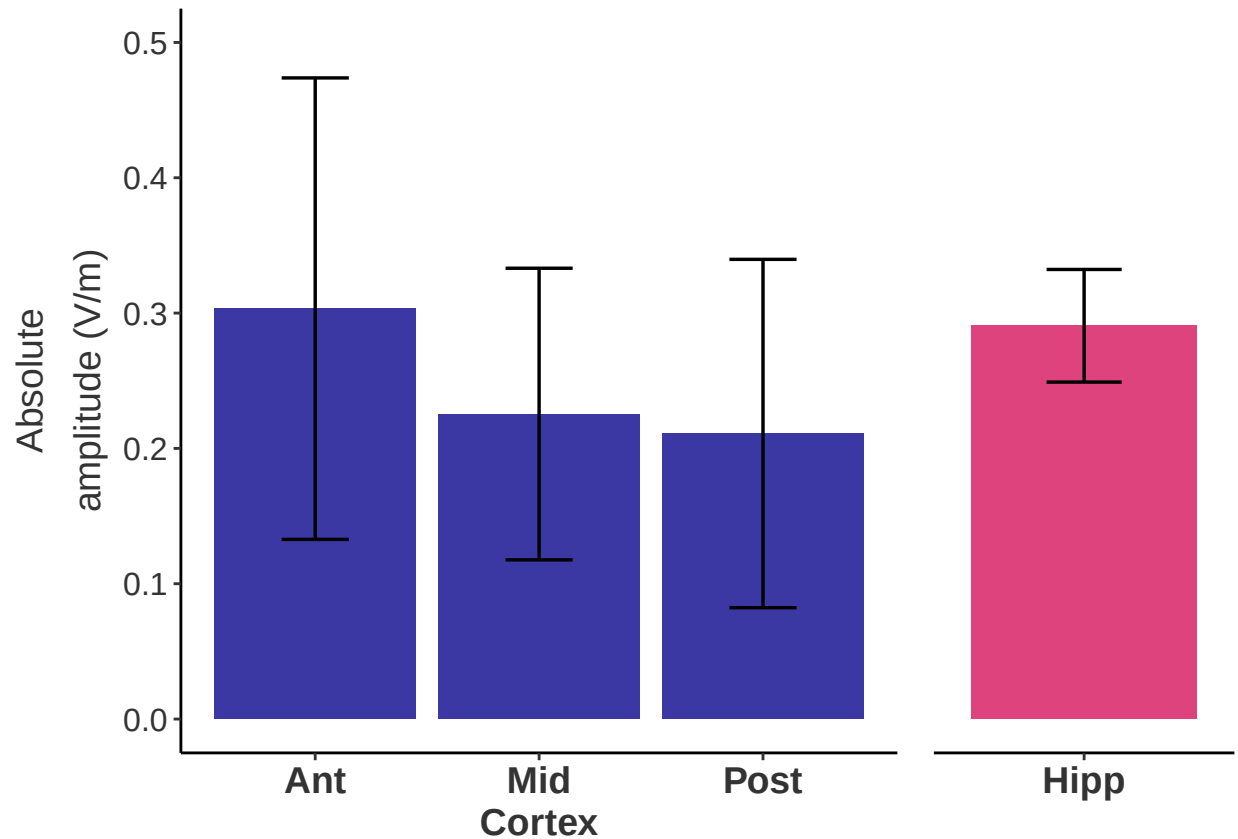
Hipp_HF = Ant_HF + Mid_HF + Post_HF
Hipp_HF_sd = Ant_HF_sd + Mid_HF_sd + Post_HF_sd
Region <- "Hipp"
Median <- Hipp_HF
SD <- Hipp_HF_sd

Thipp_HF <- data.frame(Region, Median, SD)

p.HF.field_hipp <- ggplot(Thipp_HF , aes(x = Region, y = Median)) +
  geom_bar(stat="identity", fill=c("#de437e")) +
  geom_errorbar(aes(ymin=Median-SD, ymax=Median+SD), width=0.3, size=.6, position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, .5)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_blank(),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_blank(),
    #axis.ticks.y=element_blank(),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 14), legend.position = "none") + xlab("") + easy_remove_y

Fields_cortex_hipp_HF <- grid.arrange(
  arrangeGrob(p.HF.field_cortex, p.HF.field_hipp, nrow=1, ncol=2 ,widths=c(1,0.37)))

```



```
## Normlaised to hippocampus
HF_fields_cortex <- TIfields_cortex %>% dplyr::select(ROI, MedianTEf, StdTEf)
HF_fields_cortex <- rename(HF_fields_cortex,
                           Median=MedianTEf,
                           Region = ROI,
                           SD = StdTEf)

HF_fields <- rbind(Thipp_HF, HF_fields_cortex)

HF_fields$Median[HF_fields$Region=="Hipp"]

## [1] 0.2906262

norm <- ddply(HF_fields, c("Region"), summarise,
              Mnorm = (Median*100)/(HF_fields$Median[HF_fields$Region=="Hipp"]),
              SDnorm = (SD)/(HF_fields$Median[HF_fields$Region=="Hipp"])*100
              )

norm_cortex <- subset(norm, Region!="Hipp")
p.HF.field_cortex_norm <- ggplot(norm_cortex, aes(x = Region, y = Mnorm)) +
  geom_bar(stat="identity", fill=c("#3b37a3", "#3b37a3", "#3b37a3")) +
  geom_errorbar(aes(ymin=Mnorm-SDnorm, ymax=Mnorm+SDnorm), width=0.3, size=.6,
               position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, 170)) +
```

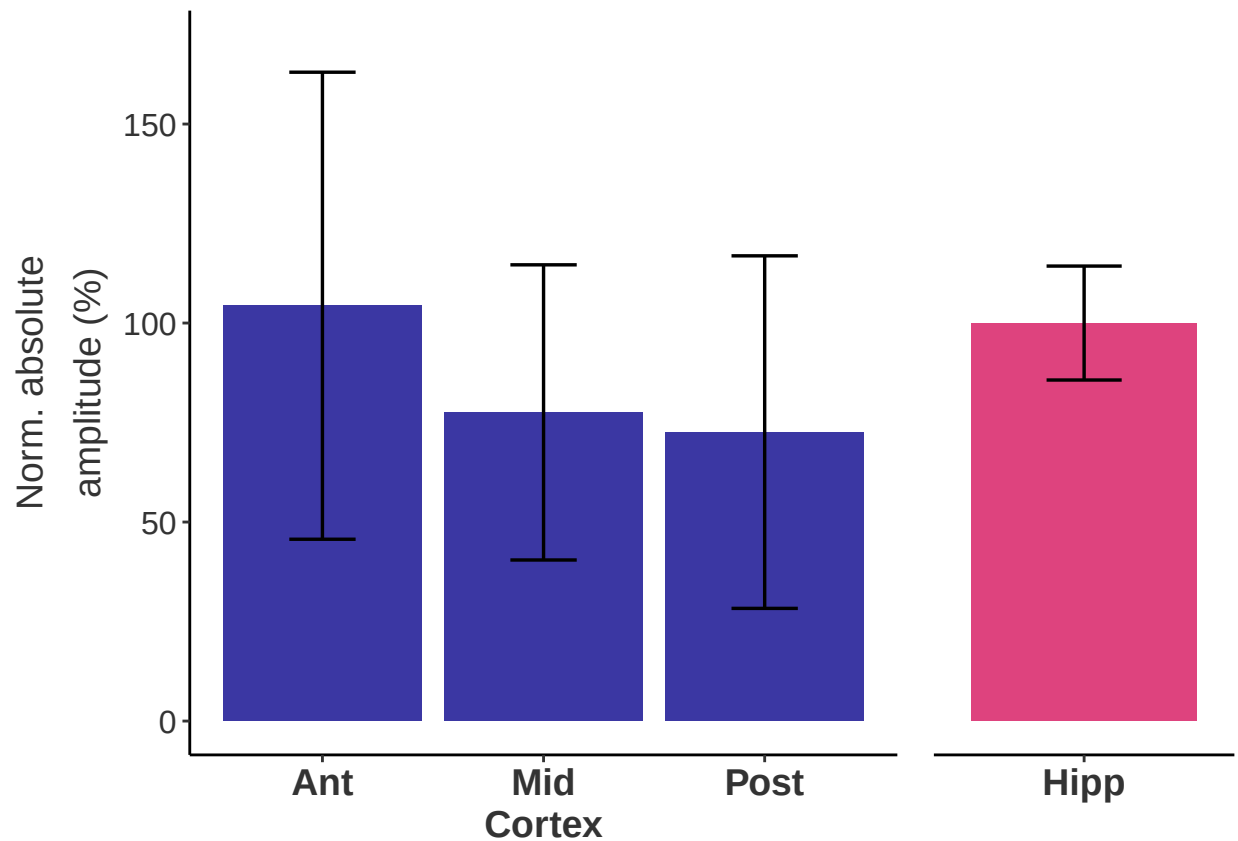
```

theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
      axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
      axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
      axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
      axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
      legend.text = element_blank(),
      legend.background = element_blank(),
      legend.key=element_blank(),
      legend.title=element_blank(),
      panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
        strip.text.y = element_text(size = 14), legend.position = "none") + ylab(expression(atop("Norm

norm_hipp <- subset(norm, Region=="Hipp")
p.HF.field_hipp_norm <- ggplot(norm_hipp , aes(x = Region, y = Mnorm)) +
  geom_bar(stat="identity", fill=c("#de437e")) +
  geom_errorbar(aes(ymin=Mnorm-SDnorm, ymax=Mnorm+SDnorm), width=0.3, size=.6, position=position_dodge(
    scale_y_continuous(
      limits=c(0, 170))) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.text.y = element_blank(),
        axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_blank(),
        #axis.ticks.y=element_blank(),
        legend.text = element_blank(),
        legend.background = element_blank(),
        legend.key=element_blank(),
        legend.title=element_blank(),
        panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
        strip.text.y = element_text(size = 14), legend.position = "none") + xlab("") + easy_remove_y

HF_Fields_cortex_hipp_norm <- grid.arrange(
  arrangeGrob(p.HF.field_cortex_norm, p.HF.field_hipp_norm, nrow=1, ncol=2 ,widths=c(1,0.37)))

```

```
ggsave(HF_Fields_cortex_hipp_norm, file="Figure_1eii.eps", width = 9, height = 8, dpi = 300, units = "cm")
```

Plot data - Figure 1f

```
# Plot fields for hippocampus
Hipp <- subset(data, Region=="Hippocampus" & Stim=="TI")

# --# Sum across regions -- Ant/Sum; Mid/Sum; Post/Sum
Ratio_Total <- ddp(Hipp, c("Ratio"), summarise,
  Ant = Median[ROI=="Ant"] / (Median[ROI=="Ant"] + Median[ROI=="Mid"] + Median[ROI=="Post"]),
  Mid = Median[ROI=="Mid"] / (Median[ROI=="Ant"] + Median[ROI=="Mid"] + Median[ROI=="Post"]),
  Post = Median[ROI=="Post"] / (Median[ROI=="Ant"] + Median[ROI=="Mid"] + Median[ROI=="Post"])
)

Ratio_long <- tidyr::gather(Ratio_Total, Regions, metric, Ant:Post, factor_key=TRUE)
Ratio_long$Regions <- as.factor(Ratio_long$Regions)

Ratio_long$Current_Ratio <- NA
Ratio_long$Current_Ratio[Ratio_long$Ratio=="1"] <- "TI 1:1"
Ratio_long$Current_Ratio[Ratio_long$Ratio=="1.5"] <- "TI 1:3"
```

```

# Add SD to table
SDs <- Hipp[, c("Ratio", "ROI", "Std")]
SDs <- SDs %>%
  rename(Regions = ROI)
total <- merge(Ratio_long,SDs,by=c("Ratio","Regions"))

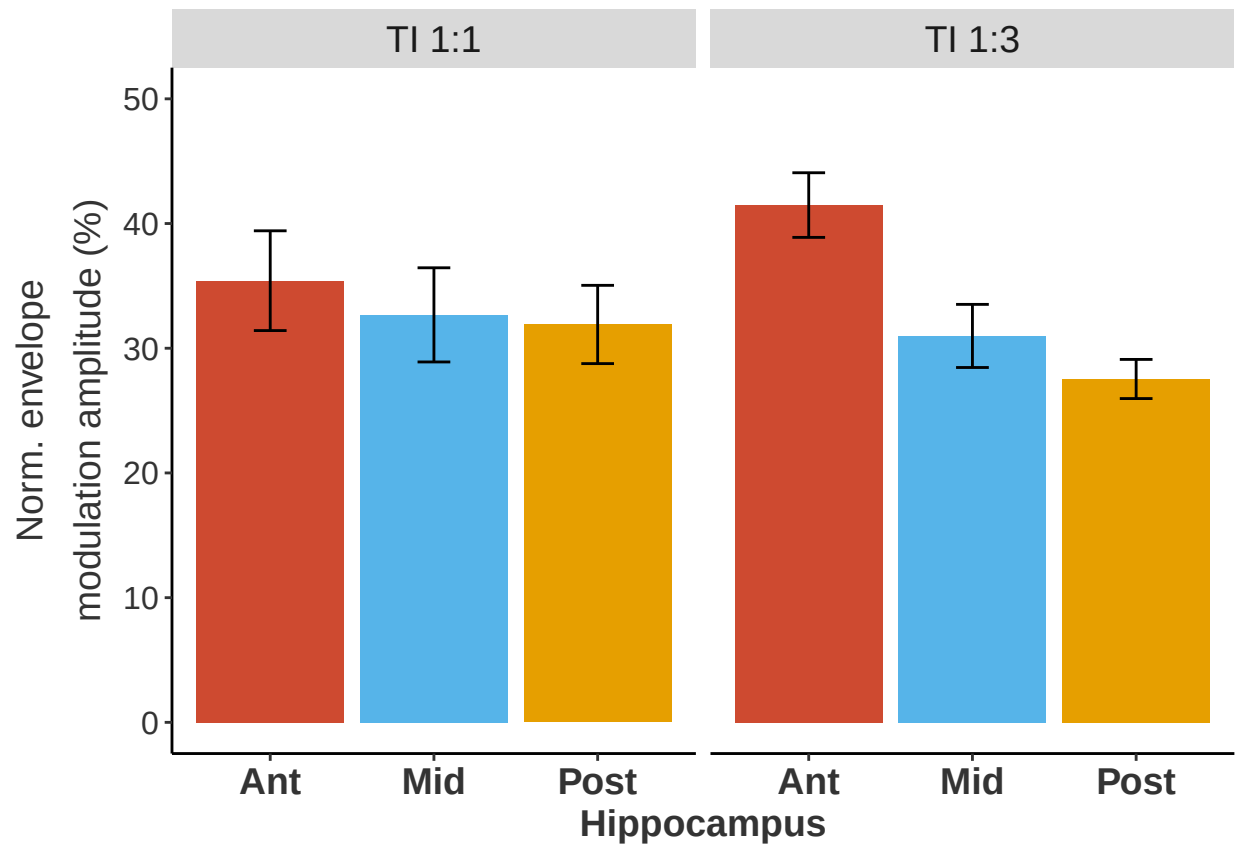
p.ratio.field <- ggplot(total, aes(x = Regions, y = metric)) +
  geom_bar(stat="identity") +
  geom_bar(stat="identity", fill=c("#ce4a30", "#56B4E9", "#E69F00","#ce4a30", "#56B4E9", "#E69F00")) +
  geom_errorbar(aes(ymin=metric-Std, ymax=metric+Std), width=.2,
    position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, .45)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 14), legend.position = "none") + xlab("Hippocampus") + ylab("Metric")

## Plot Ratios in %

p.ratio.field_prct <- ggplot(total, aes(x = Regions, y = metric*100)) +
  geom_bar(stat="identity") +
  geom_bar(stat="identity", fill=c("#ce4a30", "#56B4E9", "#E69F00","#ce4a30", "#56B4E9", "#E69F00")) +
  geom_errorbar(aes(ymin=(metric-Std)*100, ymax=(metric+Std)*100), width=.2,
    position=position_dodge(.9)) +
  scale_y_continuous(
    limits=c(0, 50)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 14), legend.position = "none") + xlab("Hippocampus") + ylab("Metric (%)")

```

```
strip.text.y = element_text(size = 14), legend.position = "none") + xlab("Hippocampus") + ylab("Norm. envelope modulation amplitude (%)")
p.ratio.field_prct
```



```
ggsave(p.ratio.field_prct, file="Figure_1f.eps", width = 9, height = 8, dpi = 300, units = "cm", device = "cairo_pdf")
```